

Part A: Complete each question, showing work in the space provided. Do not attach additional pages. Each question on this page is worth 5 pts.

Find the determinant of each matrix using any method:

1. $\begin{vmatrix} 3 & 6 & 1 \\ 0 & 6 & 4 \\ 0 & 0 & -5 \end{vmatrix}$

This is an upper triangular matrix, so the determinant is simply the product of the diagonal entries.

$$\det = (3) \cdot (6) \cdot (-5) = -90$$

3. Determine whether the points $(-2, -3)$, $(5, 4)$, $(3, 2)$ are collinear.

$$A = \begin{bmatrix} -2 & -3 & 1 \\ 5 & 4 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= -2 \begin{vmatrix} 4 & 1 \\ 2 & 1 \end{vmatrix} - (-3) \begin{vmatrix} 5 & 1 \\ 3 & 1 \end{vmatrix} + 1 \begin{vmatrix} 5 & 4 \\ 3 & 2 \end{vmatrix} \\ &= -2(4-2) + 3(5-3) + 1(10-12) \\ &= 0 \end{aligned}$$

Since the determinant is 0, the points are collinear.

2. Find the area of the triangle having the vertices $(-2, 2)$, $(3, 2)$, $(0, 4)$.

$$\text{Area} = \frac{1}{2} \det \begin{bmatrix} -2 & 2 & 1 \\ 3 & 2 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\begin{aligned} \det &= -2 \begin{vmatrix} 2 & 1 \\ 4 & 1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 3 & 2 \\ 0 & 4 \end{vmatrix} \\ &= -2(2(1) - 4(1)) - 2(3(1) - 0(1)) + 1(3(4) - 0(2)) \\ &= 4 - 6 + 12 = 10 \end{aligned}$$

$$\text{Area} = \frac{1}{2} (10) = 5$$

4. Find w such that $2\mathbf{u} + \mathbf{v} - 3\mathbf{w} = \mathbf{0}$, where $\mathbf{u} = (0, 2, -6, 0)$ and $\mathbf{v} = (0, -3, 5, 1)$

$$\begin{aligned} 3\mathbf{w} &= 2\mathbf{u} + \mathbf{v} & 2\mathbf{u} &= (0, 4, -12, 0) \\ \mathbf{w} &= \frac{1}{3}(2\mathbf{u} + \mathbf{v}) & \mathbf{v} &= (0, -3, 5, 1) \end{aligned}$$

$$2\mathbf{u} + \mathbf{v} = (0, 4, -12, 0) + (0, -3, 5, 1) = (0, 1, -7, 1)$$

$$\mathbf{w} = \frac{1}{3}(0, 1, -7, 1) = (0, \frac{1}{3}, -\frac{7}{3}, \frac{1}{3})$$

Part B: Complete each question, showing work in the space provided. Do not attach additional pages. Each question in this part is worth 10 pts.

5. Find the determinant of each matrix using any method:

$$\begin{vmatrix} 2 & 3 & 4 & 4 \\ -1 & 0 & 3 & 0 \\ 6 & 1 & -2 & 0 \\ 5 & 1 & 1 & 0 \end{vmatrix} = A$$

$$\det(A) = 4(-1)^5 \det \begin{bmatrix} -1 & 0 & 3 \\ 6 & 1 & -2 \\ 5 & 1 & 1 \end{bmatrix}$$

$$= -4 \left(-1 \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} 6 & -2 \\ 5 & 1 \end{vmatrix} + 3 \begin{vmatrix} 6 & 1 \\ 5 & 1 \end{vmatrix} \right)$$

$$= -4 \left(-1(1+2) - 0 + 3(6-5) \right)$$

$$= -4(-3 - 0 + 3)$$

$$= 0$$

6. Use the fact that matrices are row equivalent. Find a basis for the null space of A, find a basis for the row space of A, and find a basis for the column space of A.

$$A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B(\text{Row } A) = \{(1, 0, 1, 0, 1), (0, 1, -2, 0, 3), (0, 0, 0, 1, -5)\}$$

$$B(\text{Col } A) = \left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix} \right\}$$

$$B(\text{Nul } A) = \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$$

$$x_1 = -x_3 - x_5 \quad \text{Let } x_3 = s,$$

$$x_2 = 2x_3 - 3x_5 \quad \text{Let } x_5 = t$$

$$x_4 = 5x_5$$

$$\text{Then, } x_1 = -s - t \quad x = s \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix}$$

$$x_4 = 5t$$

7. Explain if S is or is not a basis for $M_{2,2}$. Be sure to check and explain both conditions!

$$S = \left\{ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

A.) Check the size:

The dimension of $M_{2,2}$ is 4

because a general 2×2 matrix has 4 free entries. Any basis for $M_{2,2}$ must therefore contain 4 matrices. But set S only contains 2 matrices, so it cannot span $M_{2,2}$.

B.) Check Linear Independence:

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Then, $c_1 A + c_2 B = 0$

$$0 \cdot c_1 + 1 \cdot c_2 = 0 \rightarrow c_2 = 0$$

$$1 \cdot c_1 + 0 \cdot c_2 = 0 \rightarrow c_1 = 0$$

Only the trivial solution happens, therefore S is linearly indep.

8. Write each vector as a linear combination of the vectors in S, if possible. Use s_1 and s_2 respectively for the vectors in the set. $S = \{(2, -1, 3), (5, 0, 4)\}$

$$a s_1 + b s_2 = a(2, -1, 3) + b(5, 0, 4) = (2a + 5b, -a, 3a + 4b)$$

a. $z = (1, -3, 5)$

$$\begin{aligned} -a &= -3 & 2a + 5b &= 1 & 3a + 4b &= 5 \\ a &= 3 & 6 + 5b &= 1 & 3(3) + 4b &= 5 \\ & & b &= -1 & 5 &= 5 \checkmark \end{aligned}$$

$$Z = 3s_1 - s_2$$

b. $v = (18, -\frac{1}{4}, \frac{59}{4})$

$$\begin{aligned} -a &= -\frac{1}{4} & 2a + 5b &= 18 & 3(\frac{1}{4}) + 4(\frac{7}{2}) &= \frac{59}{4} \\ a &= \frac{1}{4} & \frac{1}{2} + 5b &= 18 & \frac{3}{4} + \frac{56}{4} &= \frac{59}{4} \checkmark \\ & & b &= \frac{7}{2} \end{aligned}$$

$$V = \frac{1}{4}s_1 + \frac{7}{2}s_2$$

c. $w = (2, -6, 10)$

$$\begin{aligned} -a &= -6 & 2(6) + 5b &= 2 & 3(6) + 4(-2) &= 10 \\ a &= 6 & 5b &= -10 & 18 - 8 &= 10 \checkmark \\ & & b &= -2 \end{aligned}$$

$$W = 6s_1 - 2s_2$$

d. $u = (6, 1, -1)$

$$\begin{aligned} -a &= 1 & 2(-1) + 5b &= 6 & 3(-1) + 4(\frac{8}{5}) &= -1 \\ a &= -1 & 5b &= 8 & -3 + \frac{32}{5} &= -1 \\ & & b &= \frac{8}{5} & \frac{17}{5} &\neq 1 \end{aligned}$$

Not Consistent, cannot be written as a linear combination of s_1 and s_2 .

Part C: Application Questions (10 pts each). Choose 4 to solve. You may choose to solve an additional 1 question for up to 5 points of extra credit on this exam. Cross off the questions you don't want considered in your grade. Circle the question you want considered for extra credit.

For questions in Part C, I do NOT need to see row operations. Show the original matrix, then the row reduced matrix. Feel free to use an online calculator. You may also use the calculator to calculate the value of determinants in part C.

9. Find the volume of the tetrahedron having the vertices $(4, 4, -4)$, $(5, -6, -3)$, $(-6, -6, -5)$, $(0, 0, 10)$. Label your solution in cubic units.

$$V = \frac{1}{6} \left| \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix} \right|$$

$$A = \begin{bmatrix} 4 & 4 & -4 & 1 \\ 5 & -6 & -3 & 1 \\ -6 & -6 & -5 & 1 \\ 0 & 0 & 10 & 1 \end{bmatrix}$$

$$\det(A) = 1584$$

$$V = \frac{1}{6} (1584) = 264 \text{ cubic units}$$

10. The table below shows the project values (in millions of dollars) of hardback college textbooks sold in the US for the years 2007 to 2009. Create a system of linear equations for the data to find the curve $y = at^2 + bt + c$, where t is the year and $t=7$ corresponds the 2007, and y is the value of the textbooks. Use Cramer's Rule to solve the system for a , b , and c . Write the resulting polynomial equation.

Year (t)	2007	2008	2009
Value (y)	4339	4398	4483

$$2007: t = 7, y = 4339$$

$$49a + 7b + c = 4339$$

$$A = \begin{bmatrix} 49 & 7 & 1 \\ 64 & 8 & 1 \\ 81 & 9 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 4339 \\ 4398 \\ 4483 \end{bmatrix}$$

$$a = \frac{D_a}{D} = 13$$

$$2008: t = 8, y = 4398$$

$$64a + 8b + c = 4398$$

Compute determinants:

$$D = \det(A) = \det \begin{bmatrix} 49 & 7 & 1 \\ 64 & 8 & 1 \\ 81 & 9 & 1 \end{bmatrix} = -2$$

$$b = \frac{D_b}{D} = -136$$

$$2009: t = 9, y = 4483$$

$$81a + 9b + c = 4483$$

$$D_a = \det \begin{bmatrix} 4339 & 7 & 1 \\ 4398 & 8 & 1 \\ 4483 & 9 & 1 \end{bmatrix} = -26$$

$$c = \frac{D_c}{D} = 4654$$

$$D_b = \det \begin{bmatrix} 49 & 4339 & 1 \\ 64 & 4398 & 1 \\ 81 & 4483 & 1 \end{bmatrix} = 272$$

$$y = 13t^2 - 136t + 4654$$

$$D_c = \det \begin{bmatrix} 49 & 7 & 4339 \\ 64 & 8 & 4398 \\ 81 & 9 & 4483 \end{bmatrix} = -9308$$

11. Consider the differential equation $y'' + 9y = 0$ and the solutions $\{\sin 3x, \cos 3x\}$. First, verify that each solution satisfies the differential equation. Then test the set of solutions for linear independence. If the set is linearly independent, then write the general solution of the differential equation. Use C_1 and C_2 for constants.

1.) $y_1 = \sin(3x)$

$y_1' = 3\cos(3x)$

$y_1'' = -9\sin(3x)$

$y_1'' + 9y_1 = -9\sin(3x) + 9\sin(3x) = 0 \checkmark$

$y_2 = \cos(3x)$

$y_2' = -3\sin(3x)$

$y_2'' = -9\cos(3x)$

$y_2'' + 9y_2 = -9\cos(3x) + 9\cos(3x) = 0 \checkmark$

2.) $W(y_1, y_2) = \begin{vmatrix} \sin(3x) & \cos(3x) \\ 3\cos(3x) & -3\sin(3x) \end{vmatrix}$

$W = \sin(3x)(-3\sin(3x)) - \cos(3x)(3\cos(3x))$

$= -3\sin^2(3x) - 3\cos^2(3x)$

$= -3(\sin^2(3x) + \cos^2(3x))$

$= -3$

Since $W = -3 \neq 0$, the functions are linearly independent.

3.) General Solution:

Because y_1 and y_2 are linearly indep. solutions, the general

solution is: $y = C_1 \sin(3x) + C_2 \cos(3x)$

12. Perform a rotation of axes to eliminate the xy -term. Use x' and y' for the rotated coordinates.

$5x^2 - 4xy + 5y^2 - 21 = 0$

$\tan(2\theta) = \frac{B}{A-C}$ for $Ax^2 + Bxy + Cy^2$

$\tan(2\theta) = \frac{-4}{5-5} = \frac{-4}{0}$

$2\theta = \frac{\pi}{2}$

$\theta = \frac{\pi}{4}$

Since $\theta = \frac{\pi}{4}$, $x = \frac{x' - y'}{\sqrt{2}}$, $y = \frac{x' + y'}{\sqrt{2}}$

Then, $x^2 = \frac{x'^2 - 2x'y' + y'^2}{2}$, $y^2 = \frac{x'^2 + 2x'y' + y'^2}{2}$

$xy = \frac{x'^2 - y'^2}{2}$

Substitute into $5x^2 - 4xy + 5y^2 - 21 = 0$: $5\left(\frac{x'^2 - 2x'y' + y'^2}{2}\right) - 4\left(\frac{x'^2 - y'^2}{2}\right) + 5\left(\frac{x'^2 + 2x'y' + y'^2}{2}\right) - 21 = 0$

$(5-4+5)x'^2 + (5+4+5)y'^2 - 21 = 0$

$6x'^2 + 14y'^2 - 21 = 0$

$\frac{x'^2}{\frac{7}{2}} + \frac{y'^2}{\frac{3}{2}} = 1$

13. Consider $B = \{(3, 2), (1, 1)\}$, $B' = \{(-12, 0), (-4, 4)\}$ and $[x]_{B'} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

a. Find P^{-1} , the transition matrix from B to B' .

$$b_1 = (3, 2)$$

$$4b = 2 \quad -12a - 4\left(\frac{1}{2}\right) = 3$$

$$a(-12, 0) + b(-4, 4) = (3, 2)$$

$$b = \frac{1}{2} \quad a = -\frac{5}{12}$$

$$P^{-1} = P_{B' \leftarrow B} = \begin{bmatrix} -5/12 & -1/6 \\ 1/2 & 1/4 \end{bmatrix}$$

$$b_2 = (1, 1) \quad a(-12, 0) + b(-4, 4) = (1, 1) \quad 4b = 1 \quad -12a - 4\left(\frac{1}{4}\right) = 1$$

$$b = \frac{1}{4} \quad a = -\frac{1}{6}$$

b. Find P , the transition matrix from B' to B .

$$P = P_{B \leftarrow B'} = (P^{-1})^{-1}$$

$$\det(P^{-1}) = -\frac{5}{12}\left(\frac{1}{4}\right) - \left(-\frac{1}{6}\right)\left(\frac{1}{2}\right) = -\frac{5}{48} + \frac{1}{12} = -\frac{1}{48}$$

$$P^{-1} = \begin{bmatrix} -5/12 & -1/6 \\ 1/2 & 1/4 \end{bmatrix}$$

$$P = \frac{1}{\det(P^{-1})} \begin{vmatrix} d & -b \\ -c & a \end{vmatrix} \quad P = -48 \begin{vmatrix} 1/4 & 1/6 \\ -1/2 & -5/12 \end{vmatrix} = \begin{bmatrix} -12 & -8 \\ 24 & 20 \end{bmatrix}$$

c. Verify that the two transition matrices are inverses of one another.

$$PP^{-1} = I$$

$$\begin{bmatrix} -12 & -8 \\ 24 & 20 \end{bmatrix} \begin{bmatrix} -5/12 & -1/6 \\ 1/2 & 1/4 \end{bmatrix} = \begin{bmatrix} 5-4 & 2-2 \\ -10+10 & -4+5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

d. Find the coordinate matrix $[x]_B$ given the coordinate matrix $[x]_{B'}$.

$$[x]_B = P[x]_{B'} = \begin{bmatrix} -12 & -8 \\ 24 & 20 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -12 \\ 36 \end{bmatrix}$$

14. A manufacturer produces three different models of a product and ships them to two warehouses. In the matrix

$$A = \begin{bmatrix} 300 & 200 \\ 500 & 250 \\ 150 & 300 \end{bmatrix}, a_{ij} \text{ represents the number of units of model } i \text{ that the manufacturer ships to warehouse } j.$$

The matrix $B = [14.00 \quad 9.50 \quad 20.50]$ represents the prices of the three models in dollars per unit. Find the product BA and state what each entry of the matrix represents.